

Team 63: An Explainable Approach to Predicting NBA Game Winners

William Joseph Riebel III, Omer Braude, Julio Cesar Pujada Espinoza, Oluwatosin Nicholas Adegemi, Umar Rafi

1. INTRODUCTION

Sports prediction systems, including those used by betting platforms, can be very accurate, but they usually do not explain how their predictions are made. Fans and analysts are often given a single number, such as a win probability, without any insight into the reasoning behind it. This makes it difficult for people to understand or trust the model, and it limits the educational value of the prediction. In the NBA, where game outcomes depend on factors like shooting efficiency, defensive performance, momentum, rest, and overall team tendencies, a system that does not show its reasoning leaves out information that many fans would find useful.

The goal of this project is to build an explainable machine learning system for predicting NBA game winners. Instead of focusing on a complex black box model, we prioritize transparency and interpretability. Our approach uses rolling game statistics and organizes them into clear categories that fans naturally think about: Offense, Defense, Momentum, Endurance, and Hustle. We also developed an interactive web interface that visualizes these categories. The interface allows users to compare both teams, view the factors that influenced the prediction, examine category strengths, and inspect the raw statistics that contribute to each category.

By combining a clean predictive model with an easy-to-understand interface, this project aims to make NBA outcome prediction more transparent and more accessible. The system does not just estimate which team is likely to win. It also explains why the model reached that conclusion, which makes the results easier to interpret and more valuable for fans and learners.

2. PROBLEM DEFINITION

2.1 Jargon-Free Version (Heilmeier Question #1)

We are trying to build a tool that predicts the winner of an NBA game and clearly explains why one team is more likely to win than the other. Our objective is to give fans an easy way to understand the important factors behind a prediction, instead of just showing a number or percentage with no explanation. We want users to see which team has been playing better, who has more momentum, who is more rested, and how these ideas contribute to the expected outcome.

2.2 Formal Version

Given two NBA teams and their historical performance data up to the day of the matchup, predict the probability that the home team will win and produce an interpretable explanation of the factors that influenced the prediction. Let X_t represent the set of rolled statistical features available before game time t , including offensive metrics, defensive metrics, recent momentum indicators, rest-related variables, and other performance measures. Let $y_t \in \{0,1\}$ represent the true game outcome, where $y_t = 1$ indicates a home team win.

The task is to learn a predictive function

$$f: X_t \rightarrow [0,1]$$

that estimates the probability $P(y_t = 1 | X_t)$. In addition, the system must generate an interpretable explanation vector

$$E_t = g(X_t)$$

where g maps the model’s internal features to human-readable categories such as Offense, Defense, Momentum, Endurance, and Hustle. Each category score must be derived from normalized raw statistics and allow the user to understand how the values of each team compare.

The complete problem is therefore two-fold:

- (1) produce an accurate probability estimate of the game outcome, and
- (2) provide a structured and transparent explanation that shows which factors favored one team over the other. The system must present these explanations through an interactive user interface that is easy for non-experts to understand.

3. LITERATURE SURVEY

Existing research in sports prediction and explainable machine learning helps shape the design of our project. Earlier studies show that accuracy alone does not fully capture the quality of a predictive model. Walsh and Joshi [1] argue that calibration, which measures how well predicted probabilities match real outcomes, can be more informative than raw accuracy. Ren and Susnjak [11] reach a similar conclusion when evaluating explainable soccer models against bookmaker odds. These findings support our interest in probability-based evaluation methods such as Brier scores, although our project focuses more on interpretability than benchmarking against betting markets.

Research on basketball game-context variables also informs our feature design. Cuesta et al. [2] discuss home-court advantage, and Esteves et al. [3] show that back-to-back games and fatigue reduce win probability. Yan et al. [6] and Costa et al. [12] report similar patterns in professional and collegiate basketball, which supports the inclusion of rest-related and contextual variables in our model. Aoki et al. [13] highlight the role of randomness in sports outcomes, which explains why even strong models have limits and why interpretability can sometimes be more valuable than squeezing out a small gain in accuracy.

Earlier NBA prediction models [4], [5], [6] improved performance using statistical and machine-learning techniques, but most did not prioritize interpretability. More recent work by Ouyang et al. [14] and He and Choi [15] uses ensemble methods and SHAP values to improve transparency in NBA prediction. Wang et al. [16] explores explainable AI for analyzing NBA play styles and match dynamics. These models produce strong results, but their explanations often rely on technical visualizations that can be difficult for casual fans to understand. Our project extends this line of work by creating a more accessible, category-based explanation system.

Work in explainable AI for other sports also illustrates the value of transparency. Lalwani et al. [7] use explainable models for volleyball, and Chen et al. [8] apply interpretable gradient boosting to soccer. Bunker and Thabtah [9] emphasize that sports analytics needs more transparency, and Ribeiro et al. [10] introduce the popular LIME framework for model-agnostic explanations. Obradović and Kečo [17] review machine-learning approaches in sports prediction, and Odet et al. [18] apply interpretable modeling to rugby. These studies show that it is possible for accuracy and interpretability to coexist. However, most focus on a single sport or rely on complex explanation tools. Our project builds on these ideas by applying explainable modeling to NBA data at a season scale and presenting the explanations in a simplified, fan-friendly format.

4. PROPOSED METHOD

4.1 Intuition

Most existing NBA prediction models focus heavily on accuracy and use complex techniques such as deep neural networks, ensemble learning, or proprietary feature engineering. These models often perform well, but they provide little insight into why a certain team is favored. As previous work has shown, fans and analysts often do not trust predictions that feel like a black box. Even models that include explanation tools like SHAP or LIME still produce technical plots that are not intuitive for general audiences.

Our approach is designed to be more transparent and more accessible by building the explanation directly into the model pipeline. Instead of using opaque features or highly complex algorithms, we rely on rolled basketball statistics and group them into categories that fans already understand. These categories include Offense, Defense, Momentum, Endurance, and Hustle. Each category summarizes related features in a way that is simple to interpret, and the differences between the two teams map naturally to basketball knowledge.

This structure gives our model several advantages over state-of-the-art systems focused only on accuracy. First, users can clearly see how each part of a team's performance contributes to the final prediction. Second, our visual interface allows people to compare strengths and weaknesses side by side. Third, fans can access the raw data behind each category, which supports transparency and trust. Even if our accuracy is slightly lower than a more complex model, the improved interpretability makes the system more useful for casual users and more educational for students learning about sports analytics and machine learning.

Overall, the intuition behind our method is that a simpler but transparent model can provide more value than a complex but opaque one. By organizing features into understandable categories and visualizing them clearly, we allow users to reason about the prediction instead of blindly accepting it.

4.2 Our Approach

Data and Feature Engineering - We collected several seasons of NBA game data and created a rolling feature set. Instead of using single-game box scores, we constructed time-lagged features that only use information available before each game. For each team, we aggregated numeric statistics such as points scored, field goal percentages, rebounds, turnovers, and opponent metrics. We also included context-based variables such as rest days and back-to-back indicators.

To improve interpretability, we grouped features into the following five categories:

- Offense: includes scoring, shooting efficiency, assists, and shot-making indicators.
- Defense: includes opponent shooting percentages and opponent scoring metrics.
- Momentum: includes recent win rate, point differential, and form indicators.
- Endurance: includes rest days, back-to-back status, and short-term workload.
- Hustle: includes rebounds, steals, turnovers forced, and possessions gained.

For each numeric feature, we normalized values based on the minimum and maximum observed across the rolled dataset. Category scores were computed by averaging normalized feature values within each category. This method produces a score from 0 to 100 that reflects each team's relative strength.

We then constructed model-ready features by mapping home and away team statistics into a single row. This includes `home_team_feature`, `away_team_feature`, opponent features, and derived ratios when needed. This structure ensures that the model receives symmetrical information for both teams.

Model Selection - We used a machine learning classifier trained on the rolled dataset to predict the probability that the home team wins. The model outputs a probability between 0 and 1. We selected a model that balances predictive performance and explainability. While more complex methods could improve accuracy slightly, they would not increase transparency, which is the main goal of the project.

To support interpretation, we compute a confidence score based on the distance from 0.5. A value close to 0.5 indicates uncertainty, while values near 0 or 1 indicate stronger confidence.

Explanation Pipeline - The explanation pipeline was designed to show exactly why the model predicted a certain outcome. It includes the following steps:

- 1) **Category Scoring:** For each team, we compute a 0 to 100 score for Offense, Defense, Momentum, Endurance, and Hustle.
- 2) **Contribution Calculation:** We subtract `away_team_category` from `home_team_category` to produce a contribution value for each category. Positive values favor the home team, and negative values favor the away team.
- 3) **Raw Stat Extraction:** For transparency, we return the first set of raw statistics that contributed to each category. These raw numbers allow users to see the exact values behind the higher-level categories.
- 4) **Prediction Packaging:** The backend sends all this information in a single JSON response that includes win probability, category scores, contributions, raw stats, and category definitions.

Backend Architecture - We built the backend using a Flask API. The main responsibilities of the backend are:

- Loading the trained model and feature list.
- Retrieving the latest rolled row for each selected team.
- Building the feature vector expected by the model.
- Running the prediction and generating probability.
- Computing category scores and contributions.
- Returning an explanation object that the frontend can visualize.

The backend is lightweight and fast, and it allows the frontend to query predictions in real time.

User Interface and Frontend Design - The user interface was built using HTML, CSS, and D3.js for interactive visualizations. The interface is designed to be simple, clean, and easy to understand. Users begin by selecting a home team and an away team from dropdown menus. When they click "Explain This Matchup," the frontend sends a request to the backend API and displays the results.

The interface includes several explainable components:

- **Prediction Card:** shows the favored team, win probability, and a confidence bar.
- **Category Edge List:** highlights which team has advantages in different categories.
- **Radar Chart:** overlays the two teams on the five category axes to show overall strengths.
- **Contribution Bar Chart:** shows how much each category contributes to the prediction, with blue bars for the home team advantage and orange bars for the away team advantage.
- **Raw Stats Toggle:** reveals the exact statistics behind each category score.
- **Category Meaning Popup:** provides definitions of each category for users who are new to basketball analytics or machine learning.

All charts and tables update dynamically based on the selected teams. The D3 visualizations allow users to clearly understand the relationship between performance categories and the model predictions.

5. EVALUATION

5.1 Testbed and Experimental Questions

All evaluations were done using our rolled NBA dataset, the trained model, and our full Flask backend and D3.js frontend. The backend was run locally, and predictions were visualized through the browser interface.

Our experiments were designed to answer the following questions:

- 1) Does the model produce realistic win probabilities when using rolled features?
- 2) Do the category scores (Offense, Defense, Momentum, Endurance, Hustle) match real basketball intuition?
- 3) Are the category contributions aligned with the predicted winner?
- 4) Does the interface make explanations easy to understand?
- 5) Are the explanations stable when the underlying stats change slightly?
- 6) Does the system handle missing or unusual data gracefully?

These questions allowed us to evaluate both predictive behavior and interpretability.

5.2 Experiments and Observations

Experiment 1: Prediction Behavior on Known Matchups

Method: We tested the model on matchups involving strong and weak teams.

Observations:

- Strong teams with good recent performance received higher win probabilities.
- Close matchups produced confidence values near 50 percent, which felt realistic.
- Fatigue and poor momentum decreased predicted odds as expected.

Experiment 2: Category Score Consistency

Method: We compared category scores with known team tendencies.

Observations:

- High-efficiency teams scored higher in Offense.
- Defensive teams scored higher in Defense.
- Back-to-back scheduling lowered Endurance scores.
- Momentum captured winning streaks accurately.

Experiment 3: Alignment of Contributions With Predictions

Method: We checked whether category differences supported the predicted winner.

Observations:

- When the model favored the home team, at least one category usually leaned their way.
- Away-team predictions occurred when the away team led in several categories.
- Contributions gave clear reasoning for each prediction.

Experiment 4: User Interface Interpretability

Method: We interacted with the radar chart, bar chart, and raw stats table.

Observations:

- The radar chart made strength differences obvious at a glance.
- The bar chart showed which categories mattered most.
- The raw stats table increased trust by showing exact underlying numbers.
- The confidence bar was straightforward for users to understand.

Experiment 5: Sensitivity to Small Data Changes

Method: We examined category values across adjacent games.

Observations:

- Category scores changed smoothly from game to game.
- Momentum and Endurance behaved predictably as teams won or rested.
- No category showed erratic or unstable behavior.

Experiment 6: Edge Cases

Method: We evaluated predictions when some data was missing or limited.

Observations:

- The system produced neutral category scores when needed.
- Low-confidence predictions signaled uncertainty appropriately.
- The UI handled missing values without breaking.

Across all experiments, the model produced reasonable predictions and highly interpretable explanations. While it cannot match the predictive accuracy of commercial betting systems, it succeeds in its primary goal: offering a transparent, easy-to-understand breakdown of why one NBA team may be favored over another.

6. CONCLUSION AND DISCUSSION

This project introduced an explainable machine learning system for predicting NBA game winners using rolled statistical features and a transparent, user-friendly interface. Our main contribution is the emphasis on interpretability rather than raw predictive accuracy. By grouping features into intuitive categories such as Offense, Defense, Momentum, Endurance, and Hustle, and visualizing them through radar charts, bar charts, and raw stat tables, users can clearly understand why the model favors one team over another.

Although the model cannot match the accuracy of professional betting algorithms, it consistently produced reasonable win probabilities and explanations that align with real basketball intuition. The system's transparency, stable category scores, and interactive visuals demonstrate that explainability can make sports prediction more accessible to fans, students, and analysts.

There are several directions for future work. Incorporating injury reports, lineup changes, or real-time data could improve prediction quality. Additional visualization modes or user testing could further enhance clarity and usability. Exploring more advanced interpretable models may also add depth while maintaining transparency.

Overall, this project shows that an interpretable approach can provide meaningful insights into NBA game outcomes and serve as a useful educational tool for understanding data-driven sports analytics.

All team members contributed a similar amount of effort, except for Paul Kim, who did nothing.

References

- [1] C. Walsh and A. Joshi, “Machine learning for sports betting: should model selection be based on accuracy or calibration?,” arXiv preprint arXiv:2303.06021, 2023.
- [2] J. Cuesta et al., “Home advantage in professional basketball: crowd and travel effects,” *J. Sports Analytics*, vol. 10, pp. 201–213, 2024.
- [3] J. Esteves et al., “The impact of rest days on basketball performance,” *Eur. J. Sport Sci.*, vol. 21, pp. 599–610, 2021.
- [4] D. Miljković et al., “The use of data mining for basketball matches outcomes prediction,” *Proc. IEEE Symp. Information Technology in Sports*, 2010.
- [5] H. Nguyen et al., “NBA game prediction using machine learning techniques,” *Int. J. Data Sci.*, 2021.
- [6] R. Yan et al., “Determinants of home-team success in the NBA: a statistical analysis by quarters,” *Sports Analytics Rev.*, 2024.
- [7] A. Lalwani et al., “Machine learning in sports: a case study on using explainable models for predicting outcomes of volleyball matches,” arXiv preprint arXiv:2206.09258, 2022.
- [8] W. Chen et al., “Explainable gradient boosting for soccer outcome prediction,” *Data Mining in Sports*, 2022.
- [9] R. Bunker and F. Thabtah, “A machine learning framework for sport result prediction,” *Appl. Computing and Informatics*, vol. 15, no. 1, pp. 27–33, 2019.
- [10] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should I trust you?: Explaining the predictions of any classifier,” *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016.
- [11] R. Ren and A. Susnjak, “Predicting football match outcomes with explainable machine learning,” arXiv preprint arXiv:2211.15734, 2022.
- [12] T. Costa et al., “A machine learning approach using interpretable models for NCAA basketball,” *Proc. ICML Applications Conf.*, 2024.
- [13] R. Y. S. Aoki, R. M. Assunção, and P. O. S. V. de Melo, “Luck is hard to beat: the difficulty of sports prediction,” arXiv preprint arXiv:1706.02447, 2017.
- [14] J. Ouyang et al., “Integration of machine learning XGBoost and SHAP for NBA game prediction,” *Frontiers in Sports Analytics*, 2024.
- [15] L. He and J. Choi, “Stacked ensemble model for NBA game outcome prediction,” *Sci. Rep.*, vol. 15, no. 1, pp. 1–12, 2025.
- [16] Z. Wang et al., “Explainable AI techniques with application to NBA analytics,” *Neurocomputing*, vol. 520, pp. 32–41, 2022.
- [17] A. Obradović and N. Kečo, “Sports results prediction model using machine learning,” *Sci. Appl. Res. J.*, vol. 73, pp. 184–189, 2024.
- [18] F. Odet et al., “A machine learning and explainability-driven methodology: a rugby case study,” *Comput. Sports Sci.*, 2025.